Technical and Economic Proposal for the LavaRápido System

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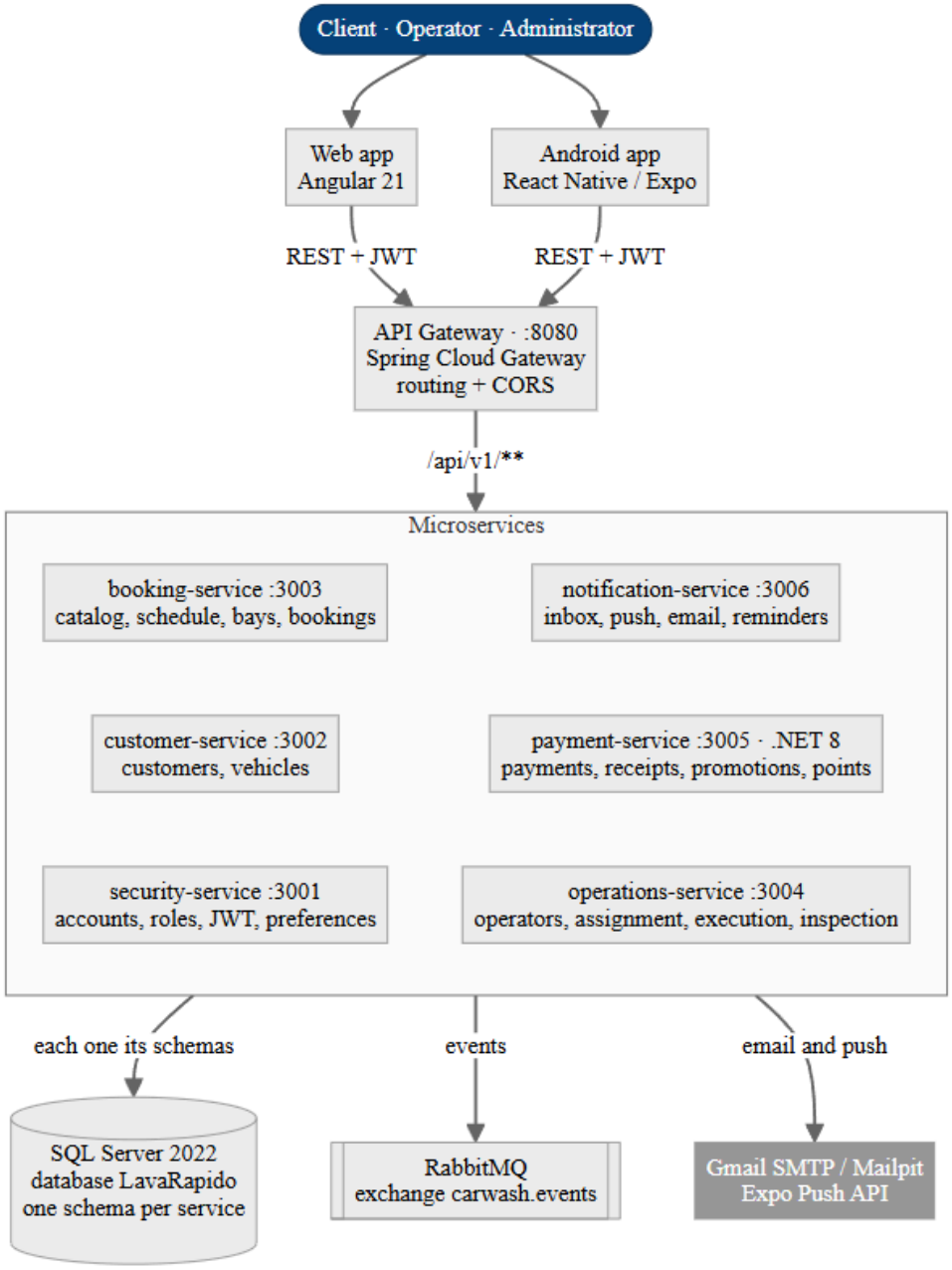
Abstract

LavaRápido is a system to manage a car wash: bookings by time slot and bay, automatic operator assignment, vehicle inspection report, digital payments with a receipt, loyalty points and notifications, with a web application and an Android application, both for the three roles. This proposal covers the technical solution (architecture, technologies and alternatives evaluated), the requirements and resources, the effort-based schedule, the team, and the cost of carrying out the project under real conditions: 28 weeks of work with go-live in week 23, a total cost of **COP 212,925,240** and a monthly operating cost of **COP 2,391,070** after launch. The problem and the objectives are in the *Software Requirements Specification* (sections 1 and 2).

1. Proposed Solution

LavaRápido consists of **six microservices** behind an API Gateway, each one owning its own part of the domain (accounts, customers, bookings, operations, payments and notifications), plus an Angular web application and a React Native Android application that use the same API. Figure 1 shows how they connect.

Figure 1 *System Component Architecture*



The REST calls between services and the RabbitMQ events are detailed in section 3.3 of the *Software Design Documentation*.

Inside each service we follow **hexagonal architecture** (domain → application → infrastructure): the business rules do not depend on Spring, JPA or the database, so they can be

tested on their own. `payment-service` follows the same idea in .NET (Clean Architecture with four projects), because it was built on a different stack; section 3 explains why.

The database is a single SQL Server instance, but each service owns its schemas and there are no foreign keys between schemas of different services: whatever a service needs from another one, it asks for through its API. That way a change in one service does not break another by accident, which is exactly the problem of a monolith with a shared database and no rules.

2. Technologies Used

Table 1 *Technologies and Their Rationale*

Layer	Technology	Why it was chosen
Backend (5 services)	Java 21 + Spring Boot 4	Booking needs strict transactional control: two customers cannot end up with the same slot and the same bay. Spring Boot already solves transaction handling, security and data access, and Java 21 brings strong typing and virtual threads to handle simultaneous bookings without blocking the server.
Backend (payment-service)	.NET 8 (ASP.NET Core, EF Core)	A separate decision, documented in ADR-013 (see section 3).
API Gateway	Spring Cloud Gateway	A single entry point on port 8080 simplifies CORS, hides the internal routes and lets us change a service internally without the web or the app noticing.
Web frontend	Angular 21	It enforces a modular structure and static typing, which helps keep the logic of three roles with different permissions organized in a single application.
Mobile frontend	React Native 0.86 + Expo SDK 57 (with expo-image-picker and expo-image-manipulator for the inspection photos)	It shares the language and the component model with the web (both in TypeScript), so the team works with a single paradigm and both applications use the same API without duplicating business logic.
Database	SQL Server 2022	The domain is highly relational (customer → vehicle → booking → operator → payment) and needs real referential integrity: a booking with two approved payments, or a receipt reused in two bookings, is not acceptable. In a document database this would have to be solved by hand in the application.
Messaging	RabbitMQ 4	Telling an operator that a service was assigned to them, or telling the customer that their payment was reviewed, should not delay the booking or payment response. With a message queue, the main operation responds right away and, if sending the notice fails, the rest of the system keeps working.
Database migrations	Liquibase	Each schema change is versioned and can be reproduced on any machine; it avoids the classic "that column exists in my database".
Authentication	JWT (HS256)	A token signed with the user's roles lets each service validate the session on its own, without asking security-service on every request.
Passwords	BCrypt	A derivation function designed to be slow on purpose, which makes brute-force attacks harder even if the database leaks.

Layer	Technology	Why it was chosen
Infrastructure	Docker + Docker Compose	The backend, the database and the messaging run the same way on any team member's computer and on the server.

3. Alternatives Considered

Every important decision was recorded as an ADR (Architecture Decision Record) in the technical documentation repository, with its context and consequences. Table 2 summarizes the alternatives that were evaluated and why they were discarded.

Table 2 *Architecture Decisions and Their Alternatives*

Decision taken	Alternatives evaluated	Why they were discarded
One schema per service in a single SQL Server instance (ADR-003)	A separate physical database for each service; a single database with no owners	A physical database per service added too much infrastructure work. A single database without ownership rules makes the boundaries between services invisible over time.
Hexagonal architecture in each service (ADR-007)	Traditional controller-service-repository layers; hexagonal only in the "core" services; CQRS with separate read models	Traditional layers mix the business rules with the framework. Applying hexagonal to only part of the system forces the team to maintain two styles. CQRS solves a scale problem this project does not have yet.
Manual payment verified by receipt, without a gateway (ADR-008)	Integrating a payment gateway (Wompi); QR without receipt review; cash only	The gateway is not part of the current data model and requires a real merchant account. A QR without review leaves no evidence that the payment happened. Accepting only cash removes digital payment, which is exactly what we wanted to offer.
`payment-service` in .NET 8 with Clean Architecture (ADR-013)	Rewriting it in Spring Boot, like the others; a single .NET project with controllers and EF Core mixed together	Rewriting it meant redoing a service that already worked just to have a uniform stack. Mixing everything in one project makes it impossible to test the business rules without depending on Entity Framework and ASP.NET.
JWT authentication and role-based authorization (ADR-006)	Stateful sessions stored on the server	A stateless token lets any service validate the session without asking another service on every request, which is simpler with microservices.
Event-driven notifications with RabbitMQ (ADR-004)	Calling the notification service over REST from each service	With a direct call, confirming a booking would depend on notification-service being available at that very moment. With events, the booking is confirmed anyway, even if the notice arrives a little later.

Note. `payment-service` is the only stack exception in the backend (.NET 8 instead of Java). It keeps the same external contract (same JWT, same error codes, same health format), so the web and the app do not notice the difference.

4. Hardware and Software Requirements

4.1 Development Team Hardware

Table 3 *Hardware per Team Member*

Equipment	Recommended specification	What it is used for
Laptop	Intel Core i5/i7 or Ryzen 5/7, 16 GB of RAM (32 GB recommended), 512 GB SSD	Running the six services, the gateway, SQL Server and RabbitMQ in Docker at the same time, together with the editor and the browser
Additional monitor (optional)	24", Full HD	Code on one screen, tests and documentation on the other
Android phone	Android 8.0 or later, with a camera	Testing the mobile app on a real device, including the photo of the receipt
Internet connection	Stable broadband	Downloading dependencies (Maven, npm, NuGet, Docker images) and working with GitHub

4.2 Development Team Software

Table 4 *Development Team Software*

Tool	Version	Use
Docker Desktop	Current	Running SQL Server, RabbitMQ, Mailpit and the backend services in containers
Java JDK	21	Building and running the five Java services and the API Gateway
.NET SDK	8	Building and running `payment-service`
Node.js	Current LTS	Building and running the web application (Angular) and the mobile application (Expo)
Visual Studio Code	Current	The team's main editor
Git	Current	Version control with GitHub
Postman	Current	Testing the API endpoints
Figma	Free plan	Wireframes and mockups

4.3 Servers

Table 5 *Test and Production Servers*

Environment	Specification	Software	Why this size
Testing (staging), W6 to W28	AWS Lightsail Linux: 2 vCPU, 8 GB of RAM, 160 GB SSD	Ubuntu Server 24.04 LTS, Docker Engine and Docker Compose, Jenkins for continuous integration and the same containers as production	Only the team, QA and the car wash (during acceptance) use it; the services run with less memory assigned and Jenkins builds one repository at a time (assumption)
Production, from W18	AWS Lightsail Linux: 4 vCPU, 16 GB of RAM, 320 GB SSD	Ubuntu Server 24.04 LTS, Docker Engine and Docker Compose, Nginx with HTTPS (Let's Encrypt), SQL Server 2022 Express, RabbitMQ 4, the six services, the gateway and the compiled web app	The six services, the gateway, SQL Server and RabbitMQ use between 8 and 10 GB of RAM under normal load (assumption); 16 GB leaves room for peaks and growth

Note. Environments planned in the proposal; the costs of section 8.2 were calculated with these specifications.

Follow-up as of October 21, 2026. For the academic demonstration, the system was deployed on a Microsoft Azure virtual machine (*Azure for Students* subscription, B2as_v2 size: 2 vCPU, 7.7 GB of RAM, and a 64 GB SSD, Ubuntu 24.04 LTS), with the eleven containers, Caddy as the HTTPS entry point, and the Android app served under */app*. With every container running it uses about 3.3 GB of RAM, so the production size of Table 5 is still comfortable. The full procedure is in the Deployment Document (folder 07). In the academic project, continuous integration was not done with Jenkins but with GitHub Actions: the 10 code repositories build and run their tests on every change, and `lavarapido-infra` also has a manual workflow that updates the server. Since October 20 and 21 the server backs up the database every day and before every update, checks each backup with a test restore and keeps the last 14; Caddy sends security headers and a content security policy; Docker checks the health of every container and caps the size of their logs. Since October 21 the API Gateway also cuts the tokens of closed sessions by asking security-service, with the answer kept 30 seconds per token, without changing the decision to use JWT.

4.4 End User

Table 6 *End-User Hardware and Software*

Role	Minimum device	Software
Any role (mobile app)	Phone with Android 8.0 or later; the customer needs a camera or gallery for the payment receipt	The app installed from Google Play
Any role (web)	Computer or laptop with an up-to-date browser	Chrome, Firefox or Edge in their current stable version
Administrator	Computer with a browser, preferably with a good resolution for the management tables	The same browser as in the previous row
Everyone	Internet connection	The system does not work offline: checking availability, booking and paying require a connection

The car wash does not need special hardware: there are no barcode readers, fiscal printers or proprietary devices. The only physical requirement is having the QR code of its digital payment accounts at hand, on screen or printed.

5. Resources: What We Already Have and What Must Be Acquired

Table 7 *Resources We Already Have*

Resource	Detail	What for
Core team	Three apprentices: project lead, backend lead and frontend lead	They cover 3 of the 7 project roles (1,920 of the 3,546 hours)
Laptops	3, one per member	Development and local testing
Internet connection	At each member's home	Remote work, dependencies and GitHub
Development tools	Visual Studio Code, JDK 21, .NET 8 SDK, Node.js, Docker Desktop, Git and Postman, installed on the three laptops	Coding, building and testing
Accounts	GitHub (project repositories) and Figma (mockups)	Code and design
Mailpit	Test mail server that runs in Docker	Testing emails without actually sending them
Knowledge of the stack	Java and Spring Boot, .NET, Angular, React Native, SQL Server, RabbitMQ and Docker	Starting construction without a learning curve

Table 8 *Resources That Must Be Acquired*

Resource	Quantity	What for	Estimated cost (COP)
Backend 2 (Java and .NET), hourly	660 h	customer-service, booking-service catalog, payment-service and notification-service	37,620,000
Mobile development (React Native), hourly	440 h	Android app for the three roles and its publication	23,320,000
QA, hourly	376 h	Test plan, cases and execution	12,220,000
DevOps, hourly	150 h	Continuous integration, servers, HTTPS, backups and monitoring	10,950,000
Test server (staging)	1, for 5.3 months	Tests by the team and QA, and acceptance by the car wash	915,800
Production server	1	Real operation of the system	824,700 during the project; 329,900 per month afterwards
Automatic backups	60 GB	Recovering the server and the database after a failure	29,500 during the project; 11,800 per month afterwards
.com.co domain	1	Web address of the system and the API	81,300 per year
HTTPS certificate (Let's Encrypt)	1	Encrypting the connection of the web and the app	0
SQL Server 2022 Express	1 instance	Production database; replaces the Developer edition of the local environment	0
Nginx	1	Reverse proxy with HTTPS and server for the compiled web app	0
Jenkins	1, on the test server	Building and running the tests of each repository on every change	0
Google Play developer account	1	Publishing the Android app	98,200 (one-time payment)
Expo account (EAS Build)	1	Building the production version of the app	0
Transactional email (Amazon SES)	3,000 emails per month	Welcome, bookings, reminders and password recovery	2,900 during the project; 1,200 per month afterwards
Uptime monitoring (UptimeRobot)	Up to 50 monitors	Alerting if the web, the gateway or any service goes down	0
Android test phone	1	Testing the app on a real entry-level device	450,000

Resource	Quantity	What for	Estimated cost (COP)
Trips to the car wash	8 visits	Requirements gathering, validations, acceptance, training and go-live	240,000
Printing	1 batch	Payment QR code for the site and training material	120,000

6. Schedule

Table 9 *Schedule by Phase*

Phase	Depends on	Weeks	Timeline	Effort (h)	Milestone at the end
Week			1···5····10···15···20···25··		
1. Kickoff and requirements	—	W1–W3	███·····	180	◆ M1 Requirements approved (W3)
2. Analysis and design	P1	W4–W6	···███·····	344	◆ M2 Design approved (W6)
3. Technical foundation	P2	W5–W7	····███·····	168	—
4. Backend construction	P3	W7–W17	·····████████·····	824	◆ M3 Backend integrated (W17)
5. Web app construction	P2, P3	W7–W18	·····███████··█·····	352	—
6. Mobile app construction	P3, P5	W10–W18	·····████████·····	328	—
7. Testing	P2, P4, P5, P6	W6–W23	·····██··████████████·····	778	◆ M4 Client acceptance (W22)
8. Deployment and go-live	P6, P7	W18–W23	···············█████·····	170	◆ M5 System in production (W23)
9. Stabilization and closure	P8	W24–W28	·····················████	228	◆ M6 Project closure (W28)
Weekly management and follow-up with the car wash	Cross-cutting	W4–W23	···████████████████·····	174	—
Total		**28 weeks**		**3,546**	

Note. W = project week; week 1 is the week the contract is signed. Each week has 5 working days of 8 hours. ▉ = week with work in the phase; ◆ = milestone. The effort adds up the hours of everyone involved.

Phase 1. Kickoff and Requirements (W1–W3)

Table 10 *Phase 1 Tasks: Kickoff and Requirements*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
1.1	Kickoff meeting with the car wash: scope, roles and work plan	—	2	W1	32	Project lead, Backend lead, Frontend lead
1.2	Requirements gathering: interviews and observation of the car wash operation	1.1	5	W1–W2	60	Project lead, Frontend lead
1.3	Requirements specification (41 RF, 12 RNF) and use cases	1.2	6	W2–W3	72	Project lead, Backend lead
1.4	Review and approval of the requirements with the client	1.3	2	W3	16	Project lead
◆ M1	**Requirements approved**	1.4	0	W3	—	—

Phase 2. Analysis and Design (W4–W6)

Table 11 *Phase 2 Tasks: Analysis and Design*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
2.1	Solution architecture and technical decisions (ADR)	1.4	5	W4	70	Backend lead, Project lead, DevOps
2.2	Data model of each service	2.1	4	W5	64	Backend lead, Backend 2
2.3	REST API and RabbitMQ event contracts	2.2	4	W5–W6	64	Backend lead, Backend 2
2.4	Wireframes and mockups in Figma (web and mobile, three roles)	1.4	10	W4–W5	100	Frontend lead, Project lead
2.5	Mockup validation with the client	2.4	1	W6	16	Frontend lead, Project lead
2.6	Test plan and acceptance criteria	1.4	3	W4	30	QA, Project lead
◆ M2	**Design approved**	2.3, 2.5, 2.6	0	W6	—	—

Phase 3. Technical Foundation (W5–W7)

Table 12 *Phase 3 Tasks: Technical Foundation*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
3.1	Repositories, branches and local environment with Docker Compose	2.1	3	W5	24	DevOps
3.2	Continuous integration with Jenkins: build and automated tests on every change	3.1	3	W5–W6	24	DevOps
3.3	Test server (staging) in the cloud	3.2	2	W6	16	DevOps
3.4	Hexagonal skeleton of the services and base migrations with Liquibase	2.3, 3.1	4	W6–W7	64	Backend lead, Backend 2
3.5	Web app foundation: structure, session, role guards and shared components	2.3, 2.5	5	W6–W7	40	Frontend lead

Phase 4. Backend Construction (W7–W17)

Table 13 *Phase 4 Tasks: Backend Construction*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
4.1	security-service: registration, sign-in and JWT issuing	3.4	6	W7–W8	48	Backend lead
4.2	security-service: roles and permissions, lockout after failed attempts, password recovery, preferences, address and users	4.1	10	W8–W10	80	Backend lead
4.3	customer-service: customers and vehicles, automatic customer creation on registration	3.4	8	W7–W8	64	Backend 2
4.4	booking-service: service catalog, opening hours, bays and business data	4.3	8	W9–W10	64	Backend 2
4.5	booking-service: bookings with slot control, statuses and cancellation	4.2, 4.4	12	W10–W12	96	Backend lead
4.6	payment-service (.NET): payment accounts with QR, payments with receipt and review	4.4	12	W10–W12	96	Backend 2
4.7	payment-service (.NET): promotions and loyalty points	4.6	8	W13–W14	64	Backend 2
4.8	operations-service: operators and their profile, shifts, absences, automatic and manual assignment, execution, ratings, metrics and inspection report	4.5	15	W13–W15	120	Backend lead
4.9	notification-service: inbox, email, push, reminders and alerts, in each account's language	4.7	13	W14–W17	104	Backend 2
4.10	API Gateway: routes, CORS and service health	4.8	3	W16	24	Backend lead
4.11	Integration between services (REST and events) and integration tests	4.9, 4.10	4	W17	64	Backend lead, Backend 2
◆ M3	**Backend integrated**	4.11	0	W17	—	—

Phase 5. Web App Construction (W7–W18)

Table 14 *Phase 5 Tasks: Web App Construction*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
5.1	Customer module: vehicles, bookings with operator choice, payments, history, ratings, points and profile	3.5	15	W7–W10	120	Frontend lead
5.2	Administrator module: catalog, opening hours, bays, business data, operators, assignment, inspection report, payments, promotions, users and roles	5.1	20	W10–W14	160	Frontend lead
5.3	Operator module: assigned services, service start and finish, ratings	5.2	5	W14–W15	40	Frontend lead
5.4	Connection to the real API through the gateway, and adjustments	5.3, 4.11	4	W18	32	Frontend lead

Phase 6. Mobile App Construction (W10–W18)

Table 15 *Phase 6 Tasks: Mobile App Construction*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
6.1	App foundation: navigation, secure session and push notification registration	5.1	5	W10–W11	40	Mobile
6.2	Customer flows: vehicles, bookings with operator choice, payment with a photo of the receipt, notifications and points	6.1	16	W11–W14	128	Mobile
6.3	Operator flows: assigned services, service start and finish	6.2	5	W14–W15	40	Mobile
6.4	Administrator module: dashboard, bookings, payments, operators, opening hours, management, reports and profile	6.3	11	W15–W17	88	Mobile
6.5	Connection to the real API, and adjustments	6.4, 4.11	4	W18	32	Mobile

Phase 7. Testing (W6–W23)

Table 16 *Phase 7 Tasks: Testing*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
7.1	Test cases traced to the requirements	2.6, 2.5	5	W6–W7	40	QA
7.2	Functional tests per module on staging, as modules are finished	7.1, 3.3, 4.3, 5.1	37	W10–W17	148	QA
7.3	Basic security and load tests (roles, tokens, simultaneous bookings)	4.11	3	W18	48	QA, DevOps, Backend lead
7.4	End-to-end system tests (web, mobile and API)	5.4, 6.5, 7.3	10	W18–W20	240	QA, Backend lead, Backend 2, Frontend lead, Mobile
7.5	Defect fixing	7.4	5	W20–W21	160	Backend lead, Backend 2, Frontend lead, Mobile, QA
7.6	Acceptance tests with the car wash	7.5	5	W21–W22	100	Project lead, QA, Frontend lead
◆ M4	**Client acceptance**	7.6	0	W22	—	—
7.7	Adjustments requested during acceptance	7.6	3	W22–W23	42	Backend lead, Backend 2, Frontend lead, Mobile

Phase 8. Deployment and Go-Live (W18–W23)

Table 17 *Phase 8 Tasks: Deployment and Go-Live*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
8.1	Production server: installation, hardening, domain and HTTPS	7.3	3	W18–W19	24	DevOps
8.2	Automatic backups, monitoring and transactional email	8.1	2	W19	16	DevOps
8.3	Google Play account, app listing and closed test (14 days)	6.5	10	W18–W20	24	Mobile, Project lead
8.4	Production version of the app and Google Play review	8.3, 7.5	5	W21–W22	10	Mobile
8.5	Training for the administrator and the operators, user manuals	7.6	3	W22–W23	36	Project lead, Frontend lead
8.6	Initial data load: catalog, opening hours, bays, operators and payment accounts	7.7, 8.2	2	W23	16	Project lead, Backend lead
8.7	Go-live	8.4, 8.5, 8.6	1	W23	44	Project lead, Backend lead, Backend 2, Frontend lead, DevOps, Mobile
◆ M5	**System in production**	8.7	0	W23	—	—

Phase 9. Stabilization and Closure (W24–W28)

Table 18 *Phase 9 Tasks: Stabilization and Closure*

ID	Task	Depends on	Duration (working days)	Weeks	Effort (h)	Responsible
9.1	Production support and incident fixing	8.7	20	W24–W27	192	Backend lead, Backend 2, Frontend lead, Project lead, Mobile, DevOps
9.2	Handover of code and documentation, and project closure	9.1	3	W28	36	Project lead, Backend lead, Frontend lead
◆ M6	**Project closure**	9.2	0	W28	—	—

7. Work Team

Table 19 *Roles, People and Dedication*

Role	Who	Responsibilities	Engagement	Weeks with dedication	Hours
Project lead	Brayan Estiven Patiño Cabrera	Requirements, work plan, weekly follow-up with the car wash, acceptance, training and closure	Core team	28 (W1–W28)	500
Backend lead	Merari Urbano Quintero	Architecture, security-service, booking-service bookings, operations-service, API Gateway and integration	Core team	28 (W1–W28)	734
Backend 2 (Java and .NET)	To be hired	customer-service, booking-service catalog, payment-service and notification-service	Hourly contract	23 (W5–W27)	660
Frontend lead	Juan Camilo Guilombo Osorio	Figma mockups, web app for the three roles and support during training	Core team	25 (W1–W28)	686
Mobile (React Native)	To be hired	Android app for the three roles and its publication on Google Play	Hourly contract	18 (W10–W27)	440
QA	To be hired	Test plan and cases, per-module, system and acceptance tests	Hourly contract	16 (W4–W22)	376
DevOps	To be hired	Local environment, continuous integration with Jenkins, servers, HTTPS, backups and monitoring	Hourly contract	10 (W4–W27)	150
Total					**3,546**

8. Project Costs

Table 20 *Calculation Assumptions*

Assumption	Value	Criterion
Working time	8 h per day, 5 working days per week, 160 h per month	Standard working hours
Hourly rate	Monthly reference salary × 1.3 ÷ 160 h, rounded	The 1.3 covers the social security and benefits paid by whoever works under an independent services contract (assumption)
Reference salaries	Lower-middle value of each national salary range	Remote hiring based in Neiva, Huila (assumption)
Exchange rate	COP 3,300 per US dollar	October 2026 official exchange rate (TRM) with a margin for fluctuation (assumption)
VAT on foreign digital services	19%	Applied to every charge in US dollars (assumption)
Contingency	10% of the direct cost	Closed scope (41 validated RF); risk in integrating six services, hiring four roles and the Google Play review (assumption)

8.1 Human Resources

Table 21 *Human Resources Cost*

Role	Hours	Weeks	Hourly rate (COP)	Cost (COP)	Criterion and source
Project lead	500	28	57,000	28,500,000	Reference of COP 7.0 M per month, the same level as a semi-senior backend developer (assumption)
Backend lead	734	28	57,000	41,838,000	Semi-senior backend in Colombia: COP 6.0–8.5 M per month (Coderhouse, 2025); 7.0 M is used
Backend 2 (Java and .NET)	660	23	57,000	37,620,000	Same range as the backend; 7.0 M is used
Frontend lead	686	25	53,000	36,358,000	Semi-senior frontend: COP 6.0–7.5 M per month (job postings on Indeed Colombia, 2026); 6.5 M is used
Mobile (React Native)	440	18	53,000	23,320,000	Remote semi-senior React Native: USD 1,600–2,750 per month (job postings on Próximo Trabajo, 2026); COP 6.5 M is used
QA	376	16	32,500	12,220,000	Semi-senior QA: COP 2.5–4.5 M per month (job postings on elempleo and Computrabajo, 2026); 4.0 M is used
DevOps	150	10	73,000	10,950,000	DevOps: COP 4.8–7 M per month in Colombia (KeepCoding, n.d.) and 8–14 M in Bogotá (Jobrise, 2026); 9.0 M is used
Total	**3,546**			**190,806,000**	

8.2 Infrastructure and Hosting

Table 22 *Infrastructure During the Project*

Item	Quantity	Assumption	Unit price	Cost (COP)	Source or criterion
Test server (staging)	1 × 5.3 months (W6–W28)	AWS Lightsail Linux: 2 vCPU, 8 GB of RAM, 160 GB SSD	USD 44 per month	915,800	AWS Lightsail pricing; billed per hour of use
Production server	1 × 2.5 months (W18–W28)	AWS Lightsail Linux: 4 vCPU, 16 GB of RAM, 320 GB SSD	USD 84 per month	824,700	AWS Lightsail pricing; from W29 it moves to the monthly operation
Automatic backups	60 GB × 2.5 months	Daily snapshots of the production server with 60 GB stored (assumption)	USD 0.05 per GB per month	29,500	AWS Lightsail snapshot pricing
.com.co domain	1 year	Yearly registration	USD 20.70	81,300	Registration price on Dynadot
HTTPS certificate	1	Let's Encrypt with automatic renewal	Free	0	Free Let's Encrypt certificates
Subtotal				**1,851,300**	

8.3 Licenses, Tools and Third-Party Services

Table 23 *Licenses, Tools and Third-Party Services*

Item	Quantity	Assumption	Unit price	Cost (COP)	Source or criterion
SQL Server 2022 Express	1 instance	Runs on the production server	Free	0	Free edition with a maximum of 10 GB per database (Microsoft); the Developer edition of the local environment cannot be used in production
Docker Engine (servers) and Docker Desktop (computers)	2 servers and 7 seats	Company with fewer than 250 employees and less than USD 10 M in annual revenue	Free	0	Docker terms of use
Java 21 (OpenJDK), Spring Boot, .NET 8, Angular, React Native and Expo SDK	—	Open-source licenses	Free	0	License of each project
RabbitMQ, Liquibase Community, Nginx, Jenkins and Ubuntu Server	—	Open-source software; Jenkins runs on the test server	Free	0	License of each project
GitHub, Figma, Visual Studio Code, Git and Postman	—	Free plans, already in use	Free	0	We already use them
Google Play developer account	1	One-time payment	USD 25	98,200	Play Console Help
App builds (Expo EAS)	Up to 15 Android builds per month	Free plan	Free	0	Expo plans
Push notifications (Expo Push)	—	Expo service with no per-message charge	Free	0	Expo plans
Transactional email (Amazon SES)	3,000 emails per month × 2.5 months	Welcome, bookings, reminders and password recovery (assumed volume)	USD 0.10 per 1,000 emails	2,900	Amazon SES pricing

Item	Quantity	Assumption	Unit price	Cost (COP)	Source or criterion
Uptime monitoring (UptimeRobot)	Up to 50 monitors, every 5 minutes	Free plan; commercial use allowed	Free	0	UptimeRobot terms
Subtotal				**101,100**	

8.4 Equipment and Other Expenses

Table 24 *Equipment and Other Expenses*

Item	Quantity	Assumption	Unit price (COP)	Cost (COP)	Source or criterion
Development laptops	3	From the core team; contractors work with their own equipment (assumption)	—	0	We already have them
Android test phone	1	Entry-level, Samsung Galaxy A06 type	450,000	450,000	Launch price in Colombia (Samsung, 2024)
Trips to the car wash	8 visits	Requirements gathering, validations, acceptance, training and go-live; local transport in Neiva (assumption)	30,000	240,000	Assumption
Printing	1 batch	Payment QR code for the site and training material (assumption)	120,000	120,000	Assumption
Subtotal				**810,000**	

8.5 Budget Summary

Table 25 *Total Project Budget*

Item	Cost (COP)
Human resources (Table 21)	190,806,000
Infrastructure and hosting (Table 22)	1,851,300
Licenses, tools and third-party services (Table 23)	101,100
Equipment and other expenses (Table 24)	810,000
Direct cost	**193,568,400**
Contingency: 10% of the direct cost (Table 20)	19,356,840
Total project cost	**212,925,240**

9. Monthly Operating Cost

Table 26 *Operation, Support and Maintenance After Closure (from W29)*

Item	Quantity	Assumption	Unit price	Monthly cost (COP)	Source or criterion
Production server	1	AWS Lightsail Linux: 4 vCPU, 16 GB of RAM, 320 GB SSD	USD 84 per month	329,900	AWS Lightsail pricing
Automatic backups	60 GB	Daily snapshots (assumption)	USD 0.05 per GB per month	11,800	AWS Lightsail pricing
Transactional email	3,000 emails	Assumed volume	USD 0.10 per 1,000 emails	1,200	Amazon SES pricing
.com.co domain	1/12 of the yearly registration	—	USD 20.70 per year	6,800	Registration price on Dynadot
HTTPS, push notifications, monitoring and builds	—	Free plans	Free	0	Tables 22 and 23
Support and maintenance	32 h	Activities in Table 27	COP 57,000 per hour	1,824,000	Backend rate (Table 21)
Operating reserve	10%	Support peaks or a larger server (assumption)	—	217,370	10% of the items above
Monthly total				**2,391,070**	
Yearly total	12 months			**28,692,840**	

10. License

The software developed in this project is licensed under an **open source with attribution** model, which allows the system to be freely distributed and modified, while keeping the obligation to acknowledge the intellectual property shared between the team and SENA.

Terms of use

1. The software may be used, copied, modified and distributed freely, as long as the authorship of the team and SENA is expressly acknowledged.

2. Any redistribution or publication of derived versions must keep the same license and attribution terms.
3. The license ensures that the software will remain open and that improvements will be available to the community under these same terms.
4. Using the software for purposes that break current law or harm the rights of the owners is prohibited.

Usage fee. Using the software for commercial or productive purposes is subject to a usage fee, whose amount is set by agreement between the team and SENA. This fee includes technical support, updates and maintenance; its reference value is the monthly operating cost in Table 26, and the payment scheme is in section 12.

Rights and responsibilities

- The team and SENA keep the ownership of the software and of their respective contributions.
- Users have the right to collaborate on and improve the system within the framework of this license.
- Any commercial or modifying use must respect these terms, ensuring attribution and the continuity of the license, as well as payment of the usage fee when applicable.

11. Maintenance Plan

Table 27 *Maintenance Activities Included in the Monthly Plan*

Activity	Frequency	Hours per month
Review of monitoring alerts, CPU and memory use, and database size against the 10 GB limit of SQL Server Express (receipts and QR codes are stored in the database as images)	Weekly	2
Backup verification and restore test	Monthly	2
Security updates for the server and for the dependencies of the services, the web and the app	Monthly	6
Fixing reported errors	As reported	12
Support for the administrator and the operators during business hours	On request	6
Compatibility with new versions of Android, the browsers and Expo SDK	As released	4
Total		**32**

Table 28 *Incident Response Times (assumption)*

Incident type	Example	First response	Expected resolution
Critical	The system does not load, or bookings and payments cannot be made	4 business hours	1 business day
Medium	A function fails without stopping the operation, for example an email that does not arrive	1 business day	3 business days
Low	Visual adjustment or usage question	2 business days	In the next update

12. Payment Schedule

Table 29 *Project Payments by Milestone*

Payment	Milestone	Week	Percentage	Amount (COP)
1	Contract signing (advance payment)	W1	30%	63,877,572
2	◆ M3 Backend integrated	W17	30%	63,877,572
3	◆ M5 System in production	W23	30%	63,877,572
4	◆ M6 Project closure	W28	10%	21,292,524
Total			**100%**	**212,925,240**

Table 30 *Payments After Go-Live*

Period	Monthly amount (COP)	What it covers
Month 1 (W24–W28)	0	Stabilization and closure, already included in the project cost (Phase 9)
From month 2	2,391,070	Server, backups, email, domain, 32 hours of support and maintenance, and operating reserve (Table 26)
Warranty, months 1 to 3	No additional cost	Fixing software errors does not use up plan hours; it is covered by the project contingency. It does not include new features

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